

Murali Krishna Akula

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PROFESSIONAL SUMMARY

Software Engineer with **3+** years of experience designing and developing cloud-native applications on **AWS**. Skilled in **Java, Spring Boot, Python, React**, and **Angular**, with expertise in microservices, secure **REST APIs**, and **CI/CD** pipelines with **Docker**. Achieved a **90%** reduction in vulnerabilities and **40%** improvement in application responsiveness

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, HTML, CSS

Libraries & Frameworks: Spring Boot, Spring, Flask, React, Angular, JUnit, Mockito

Tools & Technologies: AWS (EC2, S3, SQS, Lambda, CloudWatch), Docker, Git/GitLab, Maven, Gradle, Splunk

Databases: SQL, MySQL, PostgreSQL, Amazon DynamoDB

EXPERIENCE

Software Engineer

Feb 2025 – Present

Pointel

Edison, NJ, USA

- Established a unified error-handling mechanism for backend REST APIs across **10+** services, improving traceability and cutting debugging time by **40%**; updated Angular components to support structured error models
- Remediated **1000s** of cybersecurity vulnerabilities across **10+** repositories by fixing critical SonarQube issues, refactoring legacy code to reduce cognitive complexity by **80%**, and updating Swagger APIs for consistency

Software Engineer

Jul 2024 – Jan 2025

Intellipath Systems

Austin, TX, USA

- Designed and deployed a custom Logging Library adopted by **6** teams, eliminating sensitive data from API logs in Splunk, reducing exposure by **80%**, and standardizing logging practices across applications
- Led development of a user Billing service using Java Spring Boot and Maven; implemented JUnit 5 unit and integration tests achieving **85%** coverage, and collaborated with peers through reviews to maintain code quality
- Enhanced security by integrating Git hooks and ggshield, preventing secret leaks and reducing incidents by **90%**

Software Engineer Intern

May 2023 – Aug 2023

Viasat

Marlborough, MA, USA

- Built a full-stack web application with a Python Flask backend powering a React-based UI for **1K+** users, delivering efficient internet plan search, filtering, and visualization, and boosting user engagement by **70%**
- Optimized PostgreSQL workflows to accelerate data retrieval by **50%**, containerized backend with Docker, and automated CI/CD pipelines using GitHub Actions for streamlined development and deployment

Assistant System Engineer

Nov 2020 – Jul 2022

Tata Consultancy Services Ltd

Mumbai, MH, India

- Refactored legacy SOAP web services into modern RESTful APIs using Spring Boot and Java, increasing system performance by **40%** and facilitating faster, maintainable, and scalable applications
- Developed backend services with Java Spring Boot to power real-time agricultural data visualization for **5K+** users, working with stakeholders to deliver accurate insights on crop yield, soil moisture, and vegetation health
- Automated Python pipeline for satellite imagery, cutting processing time by **80%** & monitoring crop health faster
- Safeguarded **20+** REST APIs by introducing Role-Based Access Control using Spring Security, OAuth 2.0 authentication, and JWT-based stateless authorization, minimizing unauthorized access risks for **5K+** users

PROJECTS

Elastic Image Recognition Service | *Java, AWS, Spring Boot*

Sep 2023 – Oct 2023

- Deployed a cloud-based image recognition system on AWS (EC2, SQS, S3, CloudWatch), integrating a deep learning model in a microservices architecture with auto-scaling to process **1K+** images concurrently at **99.9%** reliability

Cloud-Based Student Attendance Recognition | *Python, AWS*

Feb 2023 – Mar 2023

- Engineered a serverless classroom attendance system using AWS Lambda, S3, and DynamoDB, containerized with Docker; auto-scaled to handle **100** classroom video uploads, reducing manual attendance tracking time by **90%**

EDUCATION

Arizona State University

Tempe, AZ, USA

Master of Science in Computer Science (GPA: 4.0)

Aug 2022 – May 2024

B V Raju Institute of Technology

Narsapur, TS, India

Bachelor of Technology in Electronics and Communication Engineering (GPA: 3.8)

Aug 2016 – May 2020